

SirenSense

Team 1: Paige Dugas, Trenton Kovach, Anthony Malone, Paul Patz, Ernest Reschke

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Project Goal

The goal of this project is to make a device that alerts drivers of an emergency vehicle siren and the direction that it was detected. This would be a useful tool for drivers that are deaf or hard of hearing.

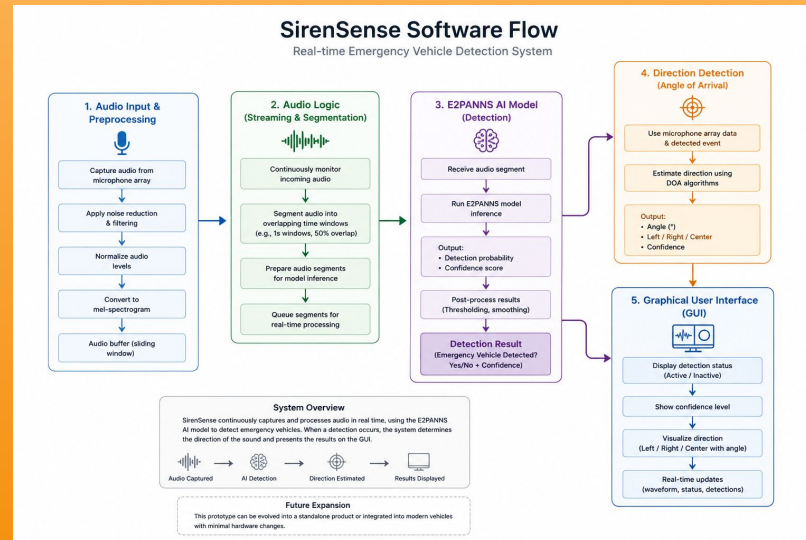
Design Requirements

With a \$500 budget we aim to build a prototype as a proof of concept of our idea of a device that can help the deaf and hard of hearing with detecting emergency vehicle sirens.

The parts must be able to ship quickly due to the limited time that the group has for the project. We also must give ourselves more compute power than we think we need for the first prototype.

Project Software

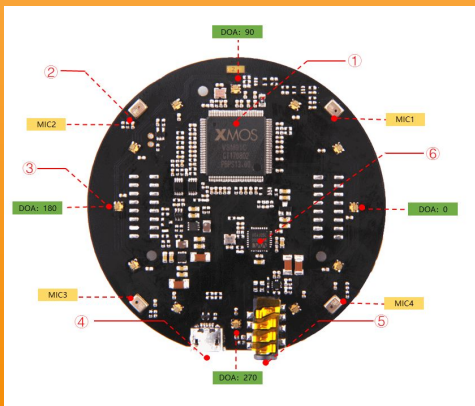
The SirenSense Project is made up of the E2PANNS AI emergency vehicle detection model, our logic to take input from our microphone and break it up into sections to send to the model for live detection. Our direction detection logic, and our graphical user interface.



Directional Auto Detection

We are now able to detect audio direction from the mic array. We have precision down to the degree.

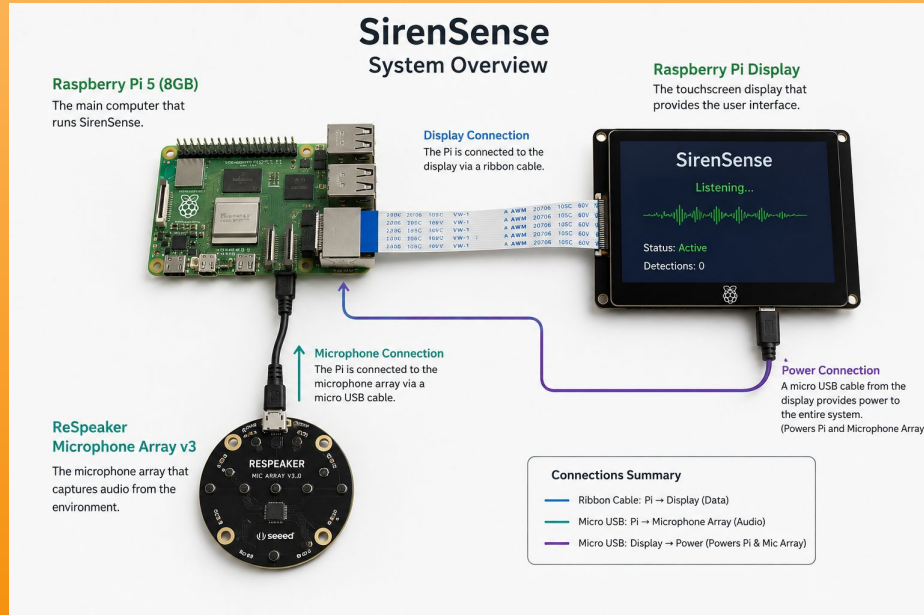
When a siren is detected, we can call a function to find the direction of arrival(DoA). This directional information can be used in conjunction with the GUI to output siren direction.



```
1  from tuning import Tuning
2  import usb.core
3  import usb.util
4  import time
5
6  dev = usb.core.find(idVendor=0x2886, idProduct=0x0018)
7
8  def sirenDetected():
9      #placeholder for siren detection logic
10     return True
11
12  def main():
13
14     if dev:
15         myTune = Tuning(dev)
16         while sirenDetected():
17             print("Siren detected at ", myTune.direction(), " degrees.")
18             time.sleep(1) #will become unnecessary when ran with siren detection
```

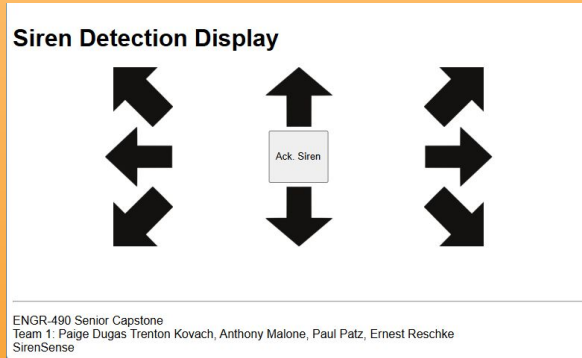
Project Electronics

The SirenSense is composed of a raspberry Pi 5 8GB, a respeaker microphone array v3, a raspberry pi display, and a cable to power it all.

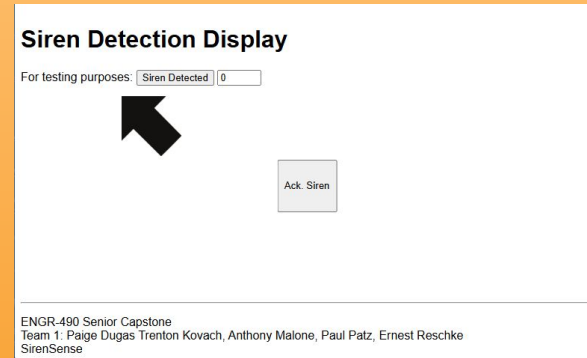


Graphic User Interface

Possible Directions:



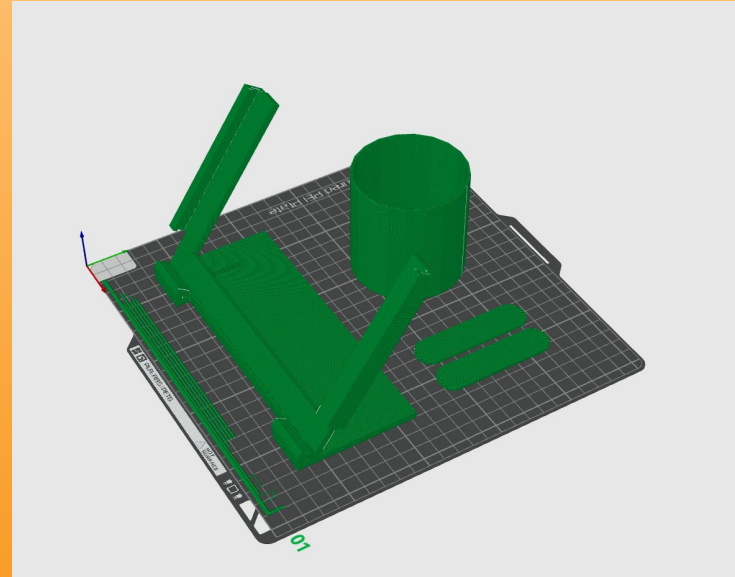
Example Display:



- The UI has 8 possible directions that can be displayed
- When a direction is displayed that means that a siren has been detected **and** the direction is identifiable by the mic array
- The acknowledge siren button can be used to hide the arrows until the display refreshes again

Enclosure

- Cup Holder has spot for legs
- Lcd screen holder will sit on top of cup
- Future iterations
 - Make lcd screen and raspberry case fully enclosed
 - Have adjustable angle to orient screen



Project Analysis

This project has gone through the following testing:

Siren Detection accuracy

How accurate is the project at detecting sirens: There was no false positive or missed sirens throughout the testing of over 100 different sirens and hundreds of other road noise and random noises.

Siren Detection Time

How quickly does the project detect a siren: It takes about 0.5 seconds to detect as that is the audio sample length and processing time needed for the model.

Software Resource Testing : The Project uses less than 3GB of resources at peak use so the 8GB raspberry pi gives us enough headroom for the rest of the project

Project Analysis cont.

This prototype is a proof of concept of what can be done to implement emergency vehicle detection into cars on the road. This can be continued by either further developing it into a standalone product or it could be integrating into modern cars with minimal physical changes needed.

If we make it a stand alone product finding a cheaper microcomputer to meet the needs for this project as well as a cheaper microphone and display could drastically reduce the price of the project to make it a more viable product.

If this were to be integrated into a modern car the only needed hardware would be a microphone that is somewhere on the outside of the car with some shielding from the wind and rain

AI Software Testing

A set of tests were done to check the softwares reliability and feasibility for this project.

Over a period of 3 hours over 156 different sounds were played and were a mix of sirens, road noise, and random sounds.

There were no false positives during the testing and a latency of less than 0.5 seconds.

After that we tested the model on the PI to confirm and validate that the model is to to big for the PI and that it was efficient enough to run without problem. The whole model uses less than 2.4GB of ram at the maximum to run. This leaves us with plenty of overhead for other parts of the project.

Future Progress



If we had more time for this project we would try to do the following

- Add volume threshold to the model to add a rough distance that the siren could be.
- Obtain a longer cable and do testing with the microphone on the outside of a car.
- Look into possible adding another model to determine what kind of emergency vehicle siren is detected.

ENGR-490 Considerations

Explain how:

A) Your engineering design consider potential public health, safety, and welfare.
Translation of stimulus into a more accessible form allows for access to more information to inform decisions and prompt reactions.

B) Your engineering design consider global, cultural, social, environmental, and economic factors.
Access to a more economical form of accessibility tools gives more opportunities for accommodation in both the personal and professional spaces.

Sources

The Project software and electronics images were created using chatgpt 5.5